Advanced Topics in Physical Chemistry

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Advanced Topics in Physical Chemistry (A26761)

Short Title:	Adv Topics in Physical Chem		
Faculty	Science and Computing		
Department	Science		
Cluster	Chemistry		
Author	CWALSH		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module covers a range of advanced topics in physical chemistry, including interfacial chemistry and polymer science. The quantum mechanics component involves an analysis of the mechanics behind standard physical chemistry problems. Heat transfer and thermodynamics concepts covered include modes of heat transfer, laws of thermodynamics, internal energy, heat, work, PV diagrams, enthalpy, entropy, Gibbs free energy and the Carnot Cycle.

Programmes

`CHEM-0033`	BSc (Hons) in Applied Chemistry with Quality Management (`WD_SQMCH_B`)	4 / 1 / M
`CHEM-0033`	BSc (Hons) in Pharmaceutical Science (`WD_SPHSC_B`)	4 / 7 / M

Indicative Content

- Physical chemistry of structured surfactant systems, including micelles, emulsions, microemulsions, gels, foams and aerosols
- Polymer science: molecular weight averages, polymer synthesis, polymer characterisation techniques, properties, industrial, pharmaceutical and biomedical applications of polymers
- Quantum mechanics: early quantum mechanical observations, photoelectric effect, particle/wave duality, DeBroglie's hypothesis, Young's double slit for electrons and waves. Heisenberg uncertainty principle. Postulates of quantum mechanics, observables and operators. Bohr atom, Schrodinger equation, wavefunctions and eigenfunctions, energy quantisation, expectation values; QM solutions of particle in a box, barrier penetration, harmonic oscillator, hydrogen atom
- Modes of heat transfer and associated problems; detailed treatment of thermodynamics; four Laws of Thermodynamics; third law and absolute entropy; heat machines, Carnot cycle and its relevance

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine the importance and use of surfactants, have an in depth knowledge of interfacial chemistry as well as related structures and processes.
2. Assess methods for polymer synthesis, relate physical and chemical properties of polymer to synthesis route and applications, characterize polymers. Discuss the use of polymers in pharmaceutical science.
3. Solve problems and write derivations based upon the laws of thermodynamics.
4. Distinguish between the three modes of heat transfer and solve associated problems.
5. Evaluate the important experiments and the Bohr model of the atom that led to the development of quantum physics.
6. Solve basic eigenvalue equations and quantum mechanical problems based on the Schrodinger equation for simple potentials, state the Heisenberg uncertainty principle and discuss the ramifications of it.

Learning and Teaching Methods

- Lectures.
- Library work.
- Set examples.
- Seminars.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	100%	1,2,3,4,5,6

Assessment Criteria

<40%: Inadequate understanding of advanced topics in physical chemistry at even a basic level. Unable to carry out critical thinking of subject area.

40%-49%: Will be able to show a basic knowledge of advanced topics in physical chemistry.

50%-59%: As well as the above, will be able to demonstrate understanding of some of the complexities of advanced topics in physical chemistry.

60%-69%: In addition, the student will demonstrate a more detailed knowledge of advanced topics in physical chemistry and demonstrate an ability to reason and analyse problems in this area.

70%-100%: All of the above to an excellent level. In addition, the learner will be able to demonstrate a comprehensive knowledge and understanding of advanced topics in physical chemistry, as well as excellent reasoning skills and focused well thought out arguments.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	48	
Independent Learning	87	

Essential Material

- "Online resources & databases." Various
- "WIT." wos.heanet.ie. www.wit.ie/library/dbs/scidbs.asp
- "WIT." www.sciencedirect.com. www.wit.ie/library/dbs/scidbs.asp
- Atkins, P. and J. De Paulo. _Physical Chemistry_. 10th ed. UK: OUP Oxford, 2014.

Supplementary Material

- Atkins, P., J. de Paulo and R. Friedman. _Quanta, Matter and Change_. 2nd ed. UK: W.H. Freeman, 2014.
- Cowie, J. and V. Arrighi. _Polymers: Chemistry and Physics of Modern Materials_. 3rd ed. USA: Taylor and Francis, 2009.
- House, J. _Fundamentals of Quantum Chemistry_. 2nd ed. London: Academic press, 2004.
- Odian, G. _Principles of Polymerization_. 4th ed. New York: Wiley, 2004.
- Rosen, M. _Surface and Interfacial Chemistry_. 4th ed. New York: Wiley, 2012.